

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application

Practical – IX

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Class:TYIT	Batch:1
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

A. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

ALGORITHM:

Install below Packages:

1. Microsoft.EntityFrameworkCore
2. Microsoft.EntityFrameworkCore.SqlServer
3. Microsoft.EntityFrameworkCore.Tools
4. Swashbuckle.AspNetCore
5. Microsoft.EntityFrameworkCore.Design

Open Visual Studio and create an **ASP.NET Core Web API** project using **.NET 8** with OpenAPI support.

Install the required **Entity Framework Core, SQL Server, Tools, Design and Swagger** packages.

Create a **Models** folder and add Student.cs with Id, Name, Age and Course properties.

Create a **Data** folder and add ApplicationDbContext.cs with DbSet<Student>.

Add the **LocalDB connection string** for StudentDB in appsettings.json.

Configure ApplicationDbContext, SQL Server and Swagger in Program.cs.

Create StudentsController and implement **GET, GET by ID, POST, PUT and DELETE** operations.

Open **Package Manager Console** and run:

- Add-Migration InitialCreate
- Update-Database

Run the project and open **Swagger UI** to test all CRUD operations.

Open **SQL Server Object Explorer** → **LocalDB** → **StudentDB** → **Tables** → **dbo.Students** and view the inserted student records.

CODE:

(StudentController.cs):

```
using Microsoft.AspNetCore.Mvc;
using Practical_9A.Data;
using Practical_9A.Models;

namespace Practical_9A.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentController : ControllerBase
```

```
{
    private readonly AppDbContext _context;

    public StudentController(AppDbContext context)
    {
        _context = context;
    }

    // GET ALL
    [HttpGet]
    public IActionResult GetResult()
    {
        return Ok(_context.Students.ToList());
    }

    // GET BY ID
    [HttpGet("{id}")]
    public IActionResult GetStudent(int id)
    {
        var student = _context.Students.Find(id);

        if (student == null)
            return NotFound();

        return Ok(student);
    }

    // INSERT
    [HttpPost]
    public IActionResult AddStudent(Student student)
    {
        _context.Students.Add(student);
        _context.SaveChanges();

        return Ok(student);
    }

    // UPDATE
    [HttpPut("{id}")]
    public IActionResult UpdateStudent(int id, Student student)
    {
        var data = _context.Students.Find(id);

        if (data == null)
            return NotFound();

        data.Name = student.Name;
        data.Age = student.Age;
        data.Course = student.Course;

        _context.SaveChanges();

        return Ok(data);
    }

    // DELETE
    [HttpDelete("{id}")]
    public IActionResult DeleteStudent(int id)
    {
        var student = _context.Students.Find(id);

        if (student == null)
            return NotFound();
    }
}
```

```
        _context.Students.Remove(student);  
        _context.SaveChanges();  
  
        return Ok();  
    }  
}
```

(Program.cs):

```
using Microsoft.EntityFrameworkCore;  
using Practical_9A.Data;  
  
var builder = WebApplication.CreateBuilder(args);  
builder.Services.AddControllers();  
  
builder.Services.AddDbContext<AppDbContext>(options =>  
    options.UseSqlServer(  
        builder.Configuration.GetConnectionString("DefaultConnection")));  
  
builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen();  
  
var app = builder.Build();  
  
app.UseSwagger();  
app.UseSwaggerUI();  
  
app.UseHttpsRedirection();  
app.UseAuthorization();  
app.MapControllers();  
  
app.Run();
```

(appsettings.json):

```
{  
  "Logging": {  
    "LogLevel": {  
      "Default": "Information",  
      "Microsoft.AspNetCore": "Warning"  
    }  
  },  
  "AllowedHosts": "*",  
  "ConnectionStrings": {  
    "DefaultConnection":  
      "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"  
  }  
}
```

(AppDbContext.cs):

```
using Microsoft.EntityFrameworkCore;  
using Practical_9A.Models;  
namespace Practical_9A.Data  
{  
    public class AppDbContext : DbContext  
    {  
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }  
        public DbSet<Student> Students { get; set; }  
    }  
}
```

(Student.cs):

```
namespace Practical_9A.Models  
{
```

```
public class Student
{
    public int Id { get; set; }
    public string Name { get; set; }
    public int Age { get; set; }
    public string Course { get; set; }
}
```

OUTPUT:

The screenshot shows a REST client interface with a dark theme. At the top, it indicates a POST request to the endpoint `/api/Student`. Below this, there are tabs for 'Parameters' and 'Request body'. The 'Parameters' tab is active, showing 'No parameters'. The 'Request body' tab is also visible, showing a JSON body: `{ "name": "Himani", "age": 20, "course": "Bsc.IT" }`. The 'Request body' tab is set to 'application/json'. At the bottom, there are 'Execute' and 'Clear' buttons.

The screenshot shows the 'Responses' section of the REST client interface. It displays the curl command used for the request: `curl -X 'POST' \ 'https://localhost:44376/api/Student' \ -H 'accept: */*' \ -H 'content-type: application/json' \ -d '{ "name": "Himani", "age": 20, "course": "Bsc.IT" }'`. The 'Request URL' is `https://localhost:44376/api/Student`. The 'Server response' section shows a 200 status code. The 'Response body' is a JSON object: `{ "Id": 1, "name": "Himani", "age": 20, "course": "Bsc.IT" }`. The 'Response headers' section shows: `content-type: application/json; charset=utf-8`, `date: Tue, 18 Aug 2020 03:25:46 GMT`, `server: Microsoft-IIS/10.0`, and `x-powered-by: ASP.NET`. At the bottom, there is a table with the response details.

Code	Description	Links
200	OK	No links

GET /api/Student/{id}

Parameters

Name
Description

id * required
integer(\$int32)
(path)

3

Execute
Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:44376/api/Student/3' \
  -H 'accept: */*'

```

Request URL

```
https://localhost:44376/api/Student/3

```

Server response

Code
Details

200

Response body

```
{
  "id": 3,
  "name": "Himani",
  "age": 20,
  "course": "Bsc.IT"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2020 03:26:33 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET

```

Responses

Code
Description

200
OK

Links

No links

PUT /api/Student/{id}

Parameters

Name
Description

id * required
integer(\$int32)
(path)

2

Request body

application/json

Request body

application/json

Edit Value | Schema

```
{
  "id": 2,
  "name": "Shubham",
  "age": 22,
  "course": "BTech.EXIT"
}
```

Execute
Clear

5

Responses

Curl

```
curl -X 'PUT' \
  'https://localhost:44376/api/Student/2' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "id": 2,
    "name": "Shubham",
    "age": 22,
    "course": "BTech.EXITC"
  }'
```

Request URL

https://localhost:44376/api/Student/2

Server response

Code	Details
200	Response body <pre>{ "id": 2, "name": "Shubham", "age": 22, "course": "BTech.EXITC" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 03:27:24 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>

Responses

Code	Description	Links
200	OK	No links

DELETE /api/Student/{id}

Parameters

Name	Description
id * required integer(\$int32) (path)	3

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
  'https://localhost:44376/api/Student/3' \
  -H 'accept: */*' \
```

Request URL

https://localhost:44376/api/Student/3

Server response

Code	Details
200	Response headers <pre>date: Tue, 18 Aug 2026 03:27:40 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>

Responses

Code	Description	Links
200	OK	No links

GET /api/Student

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:44376/api/Student' \
-H 'accept: */*'
```

Request URL

https://localhost:44376/api/Student

Server response

Code	Details						
200	Response body <pre>[{ "id": 1, "name": "Himani", "age": 20, "course": "BSc.IT" }, { "id": 2, "name": "Shubham", "age": 22, "course": "BTech.EXTC" }]</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2020 03:28:17 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre> Responses <table><thead><tr><th>Code</th><th>Description</th><th>Links</th></tr></thead><tbody><tr><td>200</td><td>OK</td><td>No links</td></tr></tbody></table>	Code	Description	Links	200	OK	No links
Code	Description	Links					
200	OK	No links					

**View → Sql Server Object Explorer → localdb → Databases → StudentDb
→ Tables → db0.Student(right click) → View Data**

dbo.Students [Data] StudentController.cs Program.cs

Max Rows: 1000

	Id	Name	Age	Course
▶	1	Himani	20	BSc.IT
	2	Shubham	22	BTech.EXTC
	4	Ritesh	21	BCA
⚙	NULL	NULL	NULL	NULL